Fifty Questions and No Answers

Machine Learning & Regularization for Financial Forecasting – the Intuition

FinTech Studio

MSc Banking and Finance

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Outline

1. Why More Predictors is Not More Knowledge
2. Leaks and the As-Of Test
3. Shrinkage: Ridge and Lasso
4. Cross-Validation in Time Order
5. The Showdown
6. Governance

From One Question to Fifty

Picture this

A student memorizes last year's exam word for word: 100% on that paper, and on this year's – new questions – worse than a classmate who guessed the average. She *memorized*; she did not *learn*. A regression with fifty predictors and a few thousand days does exactly this, and calls it "R^2".

Week 4 asked one question; this week asks fifty

- Week 4: one predictor, *yesterday's return* – mean unforecastable, variance forecastable.
- This week: ten own lags, momentum, volatility, the other assets, the day of the week, five agent proposals. Fifty predictors on pure noise earn an in-sample R^2 of about $50/2{,}500 = 2\%$ – the size of any real daily signal.

The week's thesis

A model allowed fifty predictors that **declines them all** has learned there was nothing to learn. What lets it decline: **regularization** (a price per coefficient) and **cross-validation in time order** (a rehearsal exam that sets the price).

The Rule and the Test: Delete the Future

The one rule of a feature

Every number on row t must be computable at the close of day t. The target is tomorrow's return. A feature that breaks this – a *leak* – makes the model look brilliant, and nothing in the output will tell you.

Picture this

A photograph claims to have been taken on Monday. You cannot tell by looking at it. So you ask: *if we erased everything that happened after Monday, would this photograph change?* If a face in it only appeared on Tuesday, it would. That is the **as-of test**: truncate the data at some date, rebuild every feature, and check that nothing already computed moves – and that nothing that existed disappears.

The three ways a leak gets in – all three were in the 2025 notebook

- **A window that reaches forward**: a "smoothed" centred moving average.
- **A full-sample statistic**: "high-volatility regime = above the *median*" – the median of the whole sample, 2026 included.
- **Preprocessing on everything**: a scaler fitted on the test set too.

The Agent Proposes, the Audit Decides

The course's first delegated task

We asked an AI agent for five features "computable at the close of day t". It sincerely returned a centred moving average and a median-based regime flag – both *read* like good engineering. The agent has no way to know they leak; it has never had to wait for tomorrow.

The protocol is not trust. It is audit.

- Log the prompt – it is part of the audit trail, like a trade ticket.
- Implement the proposals *exactly as written*.
- Run the as-of test. Two of five fail. Three survive and enter the model.
- Record everything in the **feature ledger**: which prompt, which proposal, which test, which verdict. The sibling of the cost ledger.

- Being legal is not being useful: the three survivors still have to earn a place in the tournament – and this week none of them does.

Overfitting, Measured in Three Numbers

OLS on fifty predictors, BTC, training sample

1. **In-sample R^2 about 3–5%.** Now scramble the target so that nothing is predictable and refit: R^2 is *still* about 2%. The number measures how many predictors you used, not what they know.
2. **Out of sample, negative.** Fit on the first 80% of the sample, score on the last 20% against the simple average: the fifty-coefficient model is *worse than a constant*.
3. **The coefficients are strangers.** The biggest one belongs to a feature nobody would have predicted, and its sign is an accident of the sample.

What a cure must do

Make the model *reluctant*: it may use a coefficient only when the data insist. Two ways to charge for reluctance follow.

Ridge: a Tax Proportional to Size

Picture this

Every coefficient pays a tax proportional to the *square* of its size. Big coefficients are expensive, so everyone shrinks – the noisiest, most uncertain ones the most. Nobody is ever taxed down to exactly zero: a small coefficient pays almost nothing, so it stays.

In one sentence of algebra

Week 3's formula with a number added to the diagonal:

$$\hat{\mathbf{w}}_{\text{ridge}} = (\mathbf{X}'\mathbf{X} + \alpha\,\mathbf{I})^{-1}\mathbf{X}'\mathbf{y}.$$

Two lines of code – the notebook does it by hand and matches scikit-learn to fifteen decimals.

- $\alpha = 0$ is plain OLS; $\alpha \to \infty$ sends everything to zero; in between, a dial.
- Ridge shrinks most where features are near-duplicates of each other – exactly where OLS is noisiest.
- **Standardize first.** A tax on size is meaningless if one coefficient is in percent and another in percent squared.

Lasso: a Fixed Toll per Feature

Picture this

Instead of a tax proportional to the square, every coefficient pays a *flat toll* proportional to its absolute size. A coefficient whose evidence is smaller than the toll cannot afford to exist at all – it is set to **exactly zero**. A larger one pays the toll and keeps the rest.

Soft-thresholding: the whole method in one picture

$$\hat{w}_{\text{lasso}} = \text{sign}(\hat{w}_{\text{OLS}}) \times \max\left(|\hat{w}_{\text{OLS}}| - \alpha,\ 0\right)$$

Below the toll, zero. Above it, moved toward zero by the toll. The absolute value has a *corner* at zero, and the optimum loves corners; the square has none.

- Lasso is a model *and* a feature-selection rule in one: the non-zero coefficients are the short list.
- Its weakness: among near-duplicate features it keeps one and drops the rest – and *which* one depends on the sample. Hence the stability check later.
- ElasticNet mixes the two tolls; the assignment's bonus.

The Coefficient Path: Watching Features Enter

Turn the dial from very expensive to free, and plot every coefficient

- **Ridge**: a fan that opens smoothly – everything grows at once, nothing is ever exactly zero.
- **Lasso**: a staircase – features enter *one at a time*, the one with the strongest evidence first, and the count of non-zero coefficients climbs from 0 to 50.

The order of entry is the data's own ranking of the predictors

On the synthetic market, the only real effect is a lag-one coefficient of 0.10 – and in one sample it does not enter first. Noise and near-duplicates out-rank the truth.

"First in" is not "true"

The path tells you *what the model would do at each price*. It does not tell you what the price should be. Only data the model has never seen can do that.

Rehearsal Exams – in the Right Order

Picture this

The final exam (the holdout) is sealed. To set the penalty you hold *rehearsal exams* inside the training data: fit on part of it, score on the rest, repeat five times, average. But a rehearsal is only fair if the questions come from *after* the material was studied.

Three rules, each a leak when broken

1. **Folds respect time.** The textbook shuffle – what the 2025 notebook did – trains on 2025 to predict 2019. The model learns the *level of risk* around each validation day from its neighbours. Use expanding windows: Week 4's protocol, five times over.
2. **Everything fitted is fitted inside the fold**, the scaler included. A `Pipeline` makes it automatic.
3. **Score against the right average.** Out-of-sample R^2 compares the model with the *training* average – never the validation average, which nobody knew at the time. Positive means "beat the free forecast"; on daily returns, half a percent is a big number.

Setting the Price, and Checking Who Keeps Winning

The curve

Plot the average rehearsal score against the penalty, with its error band. At a tiny penalty the model is OLS and loses to a constant; as the penalty grows the loss shrinks to zero – and stops there.

The one-standard-error rule

Five rehearsals give a noisy curve; the peak is partly luck. Choose the *largest* penalty – the simplest model – whose score is within one standard error of the best. For forecasting, the simpler model usually survives the real exam. Week 4's parsimony rule, with a dial instead of a parameter count.

Stability: a feature chosen once is that fold's noise

A feature counts as *stable* only if Lasso keeps it in at least four of the five rehearsals. On daily returns the stable set is empty or nearly so. That empty list is the week's quietest and most important result.

The Mean: a Tie, Read Correctly

Week 4's protocol, unchanged

Sealed holdout from 2024-10-01. One-step-ahead forecasts, model and scaler refitted every quarter, frozen in between. Benchmarks: zero and the expanding average. Scores: RMSE, MAE, R^2_{OS}. Verdict: Diebold–Mariano.

What happens

At its chosen penalty the Lasso zeros every coefficient: its forecast *is* the training average – it has **chosen the benchmark**. The scoreboard ties to the fourth decimal; DM sits in the tie zone. Ridge says the same in a whisper.

A broader negative result is a more valuable one

Week 4 found no forecastable mean with one lag; this week finds none with fifty predictors and a tuned penalty – honestly. An agent reporting "$R^2 = 0.23$" from the same inputs has leaked. The appendix shows how: a centred average, a full-sample median, a scaler on everything, shuffled folds, a penalty tuned on the holdout, R^2 against the wrong mean. Six bugs, no error message, 53% "explained".

The Variance: the Same Machine Wins

Picture this

Week 4 told the risk model what shape to be (GARCH: today's nervousness = a base level + a bit of yesterday's surprise + most of yesterday's nervousness). This week we tell the Lasso nothing. Target: tomorrow's *squared* return. Same fifty features, same rehearsals, same sealed exam.

What it finds, unprompted

- The stable set is no longer empty: recent volatility at a *short* and a *long* horizon – yesterday, last week, last month. That is the **HAR** model of realized volatility (Corsi, 2009), the industry's workhorse, discovered from data.
- Scored by QLIKE against the constant, the 21-day window and RiskMetrics' EWMA, it beats the constant with a DM statistic the mean model never produced, and competes with the practitioner's benchmarks.
- Where clustering is faint, it selects nothing and loses gracefully: an empty model is a constant, and the Risk Agent keeps GARCH.
- Returns are not forecastable; risk is. A model with no parametric form just reached Week 4's conclusion on its own.

The Analyst Agent's ML Rules v0.1

Five things the agent must prove before it may speak

Rule	In words	Why
M1 As-of test	every feature survives "delete the future", agent proposals included	a leak is invisible in the output
M2 Time-ordered tuning	folds in time order; scaler inside the fold; penalty set on training data only	a shuffled fold is an in-sample fit in disguise
M3 Beat the benchmark	$R^2_{OS} > 0$ and DM beats the average at 5% – else say "no forecastable mean"	a tie after fifty predictors is a finding
M4 Stable selection	every feature the memo names survived ≥ 4 of 5 folds	one fold's choice is one fold's noise
V1 Variance earned	the variance model beats the constant by QLIKE, DM at 5%	the Risk Agent consumes only what beat a free forecast

- The agent may build the factory, propose features, pick the penalty. The rulebook decides what it may *say*.
- In the notebook: M3 fails everywhere; V1 passes where volatility clusters loudly; M4 is the quiet rule – the memo names only what the folds agreed on, and for the mean that is nothing.
- "Our ML model explains 25% of BTC's next-day return" versus "fifty predictors, no forecastable mean; the variance model beats the constant, DM $= -2.4$": an agent that gets a bank fined versus one that gets deployed.

Key Takeaways

Before modelling

- Fifty predictors earn 2% in-sample from pure noise. Delete the future and see what changes: the as-of test on every feature, the agent's included. Log the prompt; audit the proposals.

Shrinkage

- Ridge taxes size and shrinks everyone; Lasso charges a toll and sets the unaffordable to exactly zero. Standardize first. Read the path, but let unseen data set the price.

Honest tuning and the verdict

- Rehearsals in time order, scaler inside, scored against the training average; take the simplest model within one standard error; trust only what is stable across folds. The mean: a tie, correctly read. The variance: HAR, discovered. Report both.

A model that declines fifty predictors has done its job.

Test the construction, not the result. The burden of proof is on the feature, never on the benchmark.

Thank you for your attention!

Questions and Discussion